

Lab 1: Terraform installation

Quelques faits sur Terraform

Terraform est un logiciel qui permet de gérer efficacement les ressources de l'infrastructure cloud à partir du code.

En outre, il est appelé outil «Infrastructure as Code» car il utilise un fichier de configuration pour gérer les ressources.

De plus, il prend en charge les fournisseurs de services cloud tels que AWS, Google Cloud Platform, Azure et bien d'autres.

Pour utiliser Terraform, vous devrez l'installer. HashiCorp distribue Terraform sous forme de package binaire. Vous pouvez également installer Terraform à l'aide des gestionnaires de packages courants.

Tâche 1: Installer Terraform

Installer Terraform sur CentOS

1. Installez yum-config-manager pour gérer vos référentiels.

```
sudo yum install -y yum-utils
```

2. Utilisez yum-config-manager pour ajouter le référentiel Linux officiel HashiCorp.

```
sudo yum-config-manager --add-repo https://rpm.releases.hashicorp.com/RHEL/hashicorp.repo
```

3. Installer

```
sudo yum -y install terraform
```

Installer sur Ubuntu / Debian

1. Ajouter la clé GPG HashiCorp.

```
curl -fsSL https://apt.releases.hashicorp.com/gpg | sudo apt-key add -
```

2. Ajouter le référentiel Linux officiel HashiCorp.

```
sudo apt-add-repository \  
"deb [arch=amd64] https://apt.releases.hashicorp.com $(lsb_release -cs) main"
```

3. Mettre à jour et installer

```
sudo apt-get update && sudo apt-get install terraform
```

Verifier l'installation

Enfin, vérifiez que le Terraform est capable d'accepter les commandes.

```
terraform -v
Terraform v1.3.7
on linux_amd64
```

Your version of Terraform is out of date! The latest version is 1.3.9. You can update by downloading from <https://www.terraform.io/downloads.html>

```
terraform version
Terraform v1.3.7
on linux_amd64
```

Your version of Terraform is out of date! The latest version is 1.3.9. You can update by downloading from <https://www.terraform.io/downloads.html>

```
terraform -version
Terraform v1.3.7
on linux_amd64
```

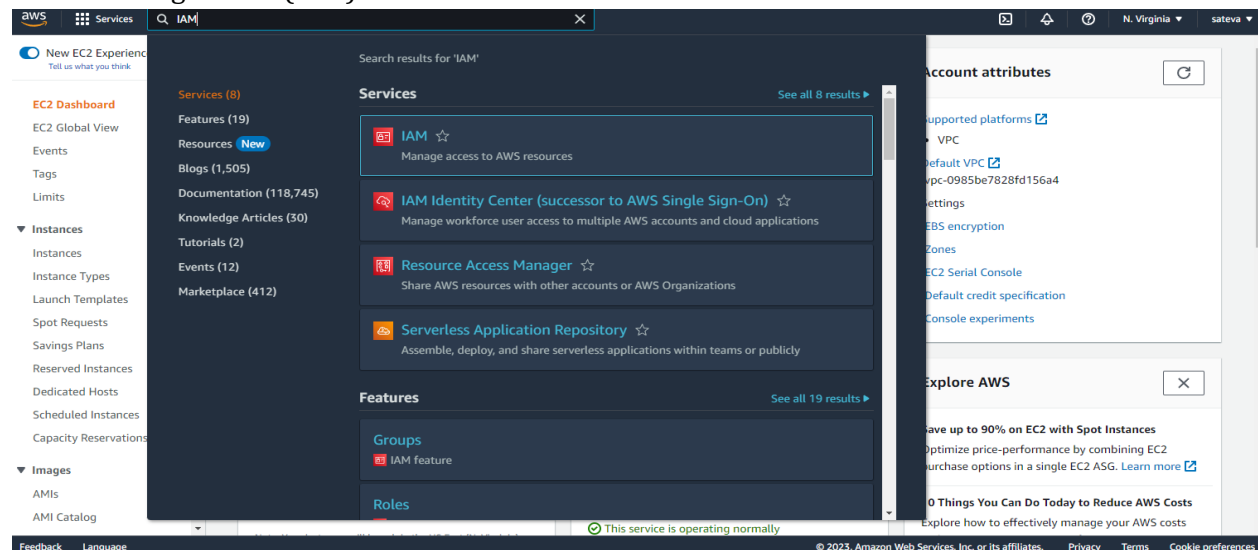
Your version of Terraform is out of date! The latest version is 1.3.9. You can update by downloading from <https://www.terraform.io/downloads.html>

Tâche 2 : Configurer votre compte AWS

Lorsque vous vous inscrivez pour la première fois à AWS, vous vous connectez initialement en tant qu'utilisateur root. Ce compte d'utilisateur a des autorisations d'accès total, donc du point de vue de la sécurité, on recommande de ne l'utiliser que pour créer d'autres comptes d'utilisateurs avec des autorisations plus limitées.

Créer un compte d'utilisateur AWS plus limité

Pour créer un compte d'utilisateur AWS plus limité, rendez-vous sur la console du service "Identity Access Management" (IAM)



Cliquez sur "Users", puis sur le bouton bleu "Add User". Saisissez un nom pour l'utilisateur. Cliquez ensuite sur le bouton Next.

Specify user details

User details

User name
terraform-aws-account

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☐ Provide user access to the AWS Management Console - optional
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

Info If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel Next

Vous arrivez ensuite sur la page de configuration des autorisations de votre nouvel utilisateur. Sélectionnez la section "Attach existing policies directly", et choisissez la policy déjà préconfigurée d'AWS nommée "AdministratorAccess". Cliquez ensuite sur le bouton Next. Ensuite Create User.

Set permissions

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1045)

Choose one or more policies to attach to your new user.

Filter distributions by text, property or value

	Policy name	Type	Attached entities
☐	AccessAnalyzerServiceRolePolicy	AWS managed	0
☒	AdministratorAccess	AWS managed - job function	2
☐	AdministratorAccess-Amplify	AWS managed	0
☐	AdministratorAccess-AWSElasticBe...	AWS managed	0
☐	AlexaForBusinessDeviceSetup	AWS managed	0

Nous venons de créer l'utilisateur terraform-aws-account. Maintenant, nous avons besoin de « l'ID de clé d'accès » et « clé d'accès secrète ».

Cliquez sur l'utilisateur.

The screenshot shows the AWS IAM console's 'Users' page. A blue banner at the top introduces the new 'Users list experience'. The left sidebar contains navigation links for 'Access management' (User groups, Users, Roles, Policies, Identity providers, Account settings) and 'Access reports' (Access analyzer, Archive rules, Analyzers, Settings, Credential report, Organization activity). The main content area shows a table of users. The user 'terraform-aws-account' is highlighted with a blue circle. The table has columns for 'User name', 'Groups', 'Last activity', 'MFA', 'Password age', and 'Active key age'.

	User name	Groups	Last activity	MFA	Password age	Active key age
☐	terraform	terraform_training	15 days ago	None	21 days ago	21 days ago
☐	terraform-aws-account	None	Never	None	None	-

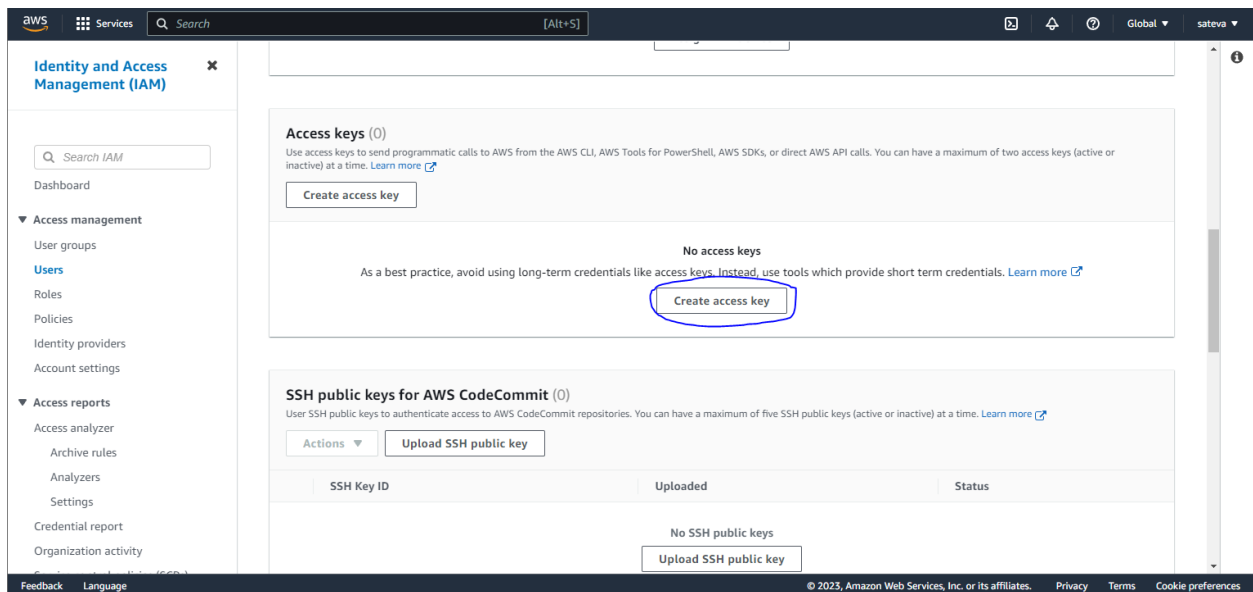
Ensuite, cliquez sur l'onglet « Security Credentials »:

The screenshot shows the 'Security Credentials' page for the 'terraform-aws-account' user. The 'Security credentials' tab is highlighted with a blue circle. The page displays a 'Summary' section with details about the user's ARN, console access status, and access keys. Below this, there are tabs for 'Permissions', 'Groups', 'Tags', 'Security credentials', and 'Access Advisor'. The 'Security credentials' tab is active, showing the 'Console sign-in' section with a link to the console and the status of the console password.

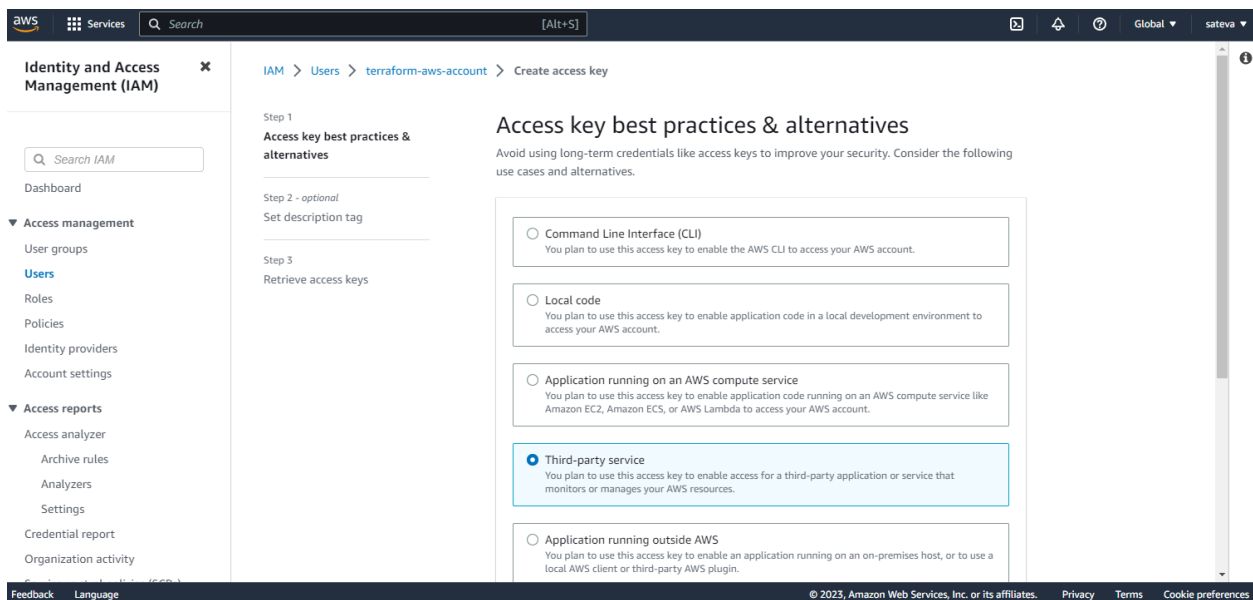
Summary		
ARN am:aws:iam::807104488271:user/terraform-aws-account	Console access Disabled	Access key 1 Not enabled
Created February 26, 2023, 18:18 (UTC+01:00)	Last console sign-in -	Access key 2 Not enabled

Console sign-in	
Console sign-in link https://807104488271.signin.aws.amazon.com/console	Console password Not enabled

Dans la section Access Keys, cliquez sur « Create Access Key » :



Choisissez « Third-party service », cochez « I understand the above recommendation and want to proceed to create an access key », next et create access key



Maintenant, nous avons la clé d'accès et la clé secrète pour la configuration du provider sur Terraform.

The screenshot displays the AWS IAM console interface. At the top, a green banner states: "This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time." The breadcrumb navigation shows the path: IAM > Users > terraform-aws-account > Create access key. The left sidebar contains the "Identity and Access Management (IAM)" menu with options like Dashboard, Access management, Roles, Policies, and Access reports. The main content area is titled "Retrieve access keys" and includes a "Step 3" section. A table displays the generated access keys:

Access key	Secret access key
AKIA3X2Z2NENH6RRGBFSW	***** Show

Below the table, the "Access key best practices" section lists several guidelines:

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

A link to "Best practices for managing AWS access keys" is provided at the bottom of the list. The footer of the console includes "Feedback", "Language", and copyright information for Amazon Web Services, Inc. or its affiliates, along with links to "Privacy", "Terms", and "Cookie preferences".